

Optimization of Multi-Stage Fuel Supply Chains under Operational and Market Constraints

Final Project – Optimization I (MATH 535)

Ignacio Conde
Illinois Institute of Technology
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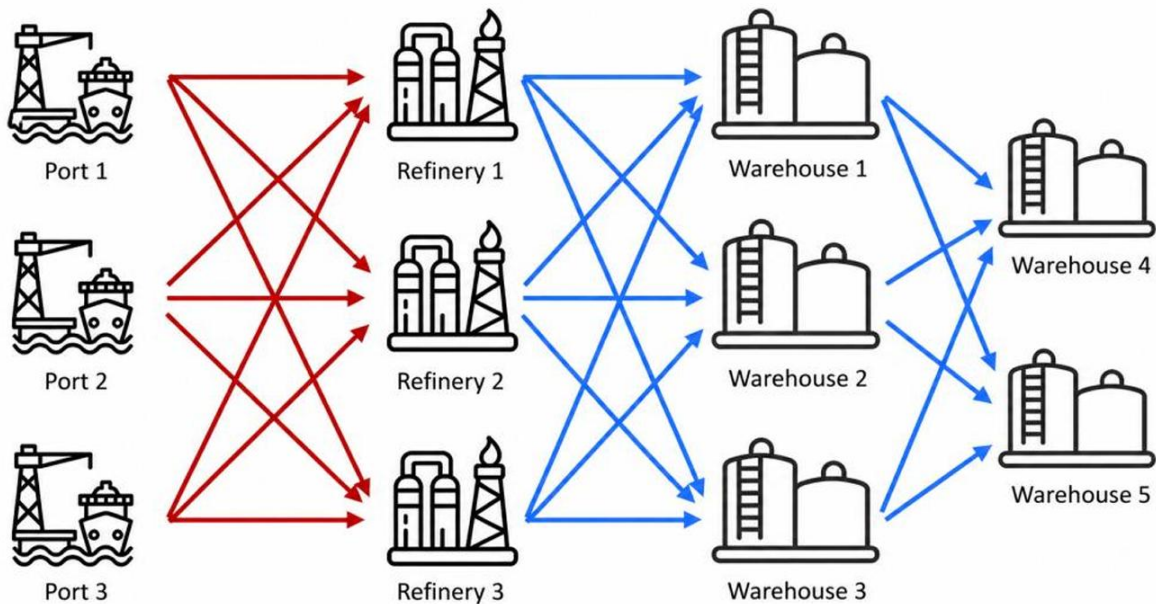


Figure 1: Multi-stage fuel supply chain network structure.

Abstract

This project formulates and solves a multi-stage fuel supply chain optimization problem in which crude oil must be transported from ports to refineries, converted into gasoline, and distributed through intermediate warehouses to final demand nodes. The case is motivated by a logistics setting in which the country lacks a pipeline network, so routing and processing decisions must be optimized through a transportation network. The model minimizes total system cost, including crude oil transportation, refinery operating costs, variable refining costs, gasoline transportation, and selected market incentives. The base model is a linear transshipment formulation, while fixed refinery activation and company assignment decisions are modeled through binary variables. For that reason, the implementation is a Mixed-Integer Linear Programming extension of the linear programming tools studied in class. Several scenarios are evaluated, including a refinery shutdown, a port supply limitation, private carrier assignment, demand growth, transportation cost inflation, and an emissions cap. The results show that the model is useful not only for finding a minimum-cost plan, but also for identifying which operational and market constraints materially change the structure and cost of the supply chain.

1. Problem Statement

The objective of this project is to determine the least-cost distribution plan for a fuel supply chain that transforms crude oil into gasoline and delivers it to final warehouses. The network has four stages: ports, refineries, intermediate warehouses, and final demand nodes. Crude oil enters the system through ports, is processed at refineries, and then moves as gasoline through the warehouse network.

The country in which the project is based lacks a mature pipeline system. This makes truck or road-based transportation a relevant operational decision rather than a secondary detail. The optimization model therefore chooses both the amount of fuel routed through each arc and, in the extended scenarios, which refineries or companies should be used.

1.1 Network and input data

The network uses the following node sets:

Set	Elements	Meaning
P	A, B, C	Ports where crude oil enters the system
R	u, v, w	Refineries where crude oil is converted into gasoline
I	1, 2, 3	Intermediate gasoline warehouses
F	4, 5	Final gasoline demand nodes
C	Company 1, Company 2	Private carriers for gasoline transportation scenarios

Table 1: Main sets used in the optimization model.

The main demand and conversion assumptions are summarized below. All flow quantities are expressed in kiloliters (kl), and all cost values in the model are expressed in thousand euros (k€).

Warehouse	1	2	3	4	5
Demand [kl]	400	300	150	250	100

Table 2: Gasoline demand by warehouse.

Parameter	Value	Unit	Interpretation
Maximum crude oil availability	3,600	kl/month	Upper bound on crude oil entering the system
Conversion factor	3:1	crude:gasoline	Every 3 kl of crude oil produce 1 kl of gasoline
Total base gasoline output	1,200	kl/month	Gasoline produced when all 3,600 kl of crude are processed

Table 3: Core model assumptions.

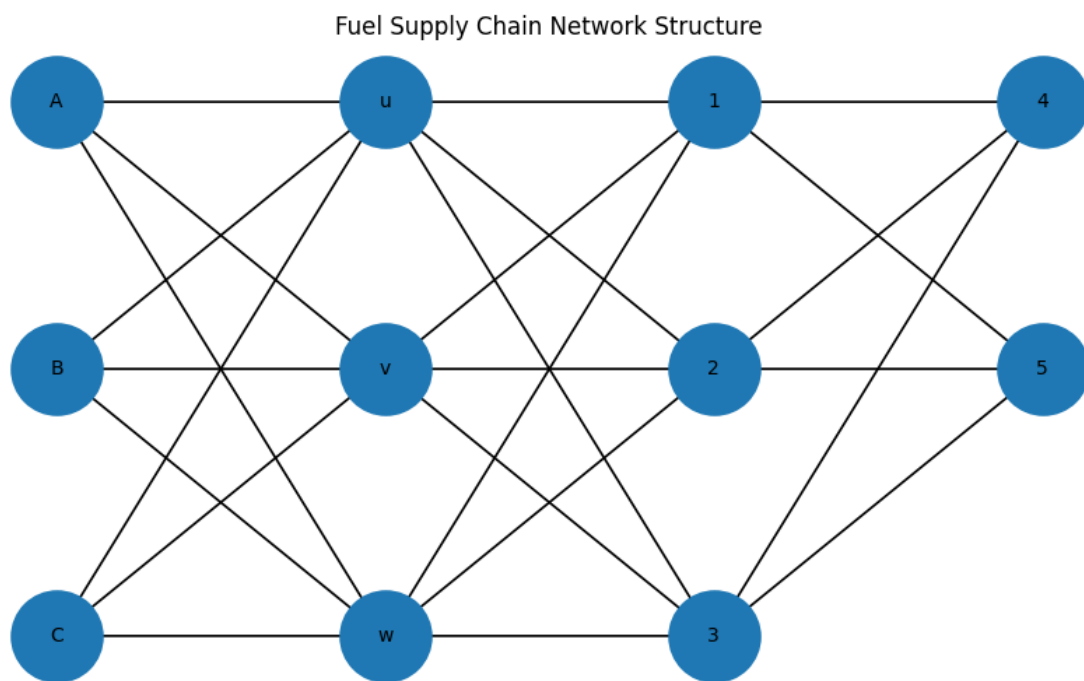


Figure 2: Complete network structure before optimization. All feasible arcs are shown.

1.2 Scenarios evaluated

The project is organized around a base case and a set of operational and market scenarios. The goal is not only to obtain one optimal distribution plan, but also to understand how different constraints affect the objective value and the selected routes.

Scenario	Model change	Purpose
Base case	Original network and costs	Benchmark minimum-cost plan
Refinery w shutdown	Refinery w cannot receive crude or ship gasoline	Operational disruption
Port C limit	Crude flow from port C limited to 1,500 kl	Supply-side restriction
Single private carrier	All gasoline arcs must be assigned to one company	Outsourcing with one carrier
Route-by-route carriers	Each gasoline arc can be assigned independently	More flexible outsourcing
Company 1 bonus	Company 1 offers a 100 k€ bonus per assigned route	Market incentive analysis
Demand +10%	Final warehouse demand increases by 10%	Demand shock
Gas transport +15%	Gasoline transportation costs increase by 15%	Cost inflation
Emissions cap 98%	Total emissions capped at 98% of the base reference	Sustainability feasibility test

Table 4: Scenario definition.

2. Mathematical Formulation of the Optimization Problem

The core model is a linear transshipment problem. Flow variables are continuous and nonnegative, the objective function is linear, and the flow-balance constraints are linear. Binary variables are added only where the real setting requires yes-or-no decisions, such as refinery activation or private carrier assignment. This makes the implemented model a Mixed-Integer Linear Programming (MILP) extension of the linear programming framework studied in class.

2.1 Decision variables

Variable	Domain	Meaning
$x_{\{p,r\}}$	$x_{\{p,r\}} \geq 0$	Crude oil shipped from port p to refinery r
$y_{\{r,i\}}$	$y_{\{r,i\}} \geq 0$	Gasoline shipped from refinery r to intermediate warehouse i
$z_{\{i,f\}}$	$z_{\{i,f\}} \geq 0$	Gasoline shipped from intermediate warehouse i to final warehouse f
δ_r	$\delta_r \in \{0,1\}$	1 if refinery r is activated; 0 otherwise
$\mu_{\{a,c\}}$	$\mu_{\{a,c\}} \in \{0,1\}$	1 if gasoline route a is assigned to company c; 0 otherwise

Table 5: Main decision variables.

2.2 Objective function

The model minimizes total system cost:

$$\min Z = \sum_{p \in P} \sum_{r \in R} c_{p,r}^{\text{crude}} x_{p,r} + \sum_{r \in R} F_r \delta_r + \sum_{p \in P} \sum_{r \in R} c_r^{\text{ref}} x_{p,r} + \sum_{r \in R} \sum_{i \in I} c_{r,i}^{\text{gas}} y_{r,i} + \sum_{i \in I} \sum_{f \in F} c_{i,f}^{\text{gas}} z_{i,f} - \sum_{a \in A} b_a \mu_{a, \text{Company 1}}$$

The first term captures crude oil transportation, the second fixed refinery costs, the third variable refining costs, the fourth and fifth gasoline transportation costs, and the last term the bonus offered by Company 1 in the incentive scenario. In scenarios without private carrier bonuses, the final term is omitted.

2.3 Core constraints

Crude oil shipped from each port cannot exceed its available supply:

$$\sum_{r \in R} x_{p,r} \leq S_p \forall p \in P$$

Total crude oil used cannot exceed the maximum monthly availability:

$$\sum_{p \in P} \sum_{r \in R} x_{p,r} \leq S^{\max}$$

A refinery can only receive crude oil if it is active:

$$\sum_{p \in P} x_{p,r} \leq M \delta_r \forall r \in R$$

The conversion process imposes that every 3 kl of crude oil generate 1 kl of gasoline:

$$\sum_{i \in I} y_{r,i} = \frac{1}{3} \sum_{p \in P} x_{p,r} \forall r \in R$$

For each intermediate warehouse, gasoline inflow must equal gasoline outflow:

$$\sum_{r \in R} y_{r,i} = \sum_{f \in F} z_{i,f} \forall i \in I$$

Each final warehouse must receive exactly its demand:

$$\sum_{i \in I} z_{i,f} = D_f \forall f \in F$$

2.4 Scenario-specific constraints

Refinery shutdown:

$$\delta_w = 0, x_{p,w} = 0 \forall p \in P, y_{w,i} = 0 \forall i \in I$$

Port C limitation:

$$\sum_{r \in R} x_{c,r} \leq 1500$$

Private carrier assignment:

$$\sum_{c \in C} \mu_{a,c} = 1 \forall a \in A$$

$$q_{a,c} \leq M \mu_{a,c} \forall a \in A, c \in C$$

Demand shock:

$$D_f^{\text{shock}} = 1.10 D_f \forall f \in F$$

Emissions cap:

$$\sum \text{emissions from crude transport, refining, and gasoline transport} \leq E^{\max}$$

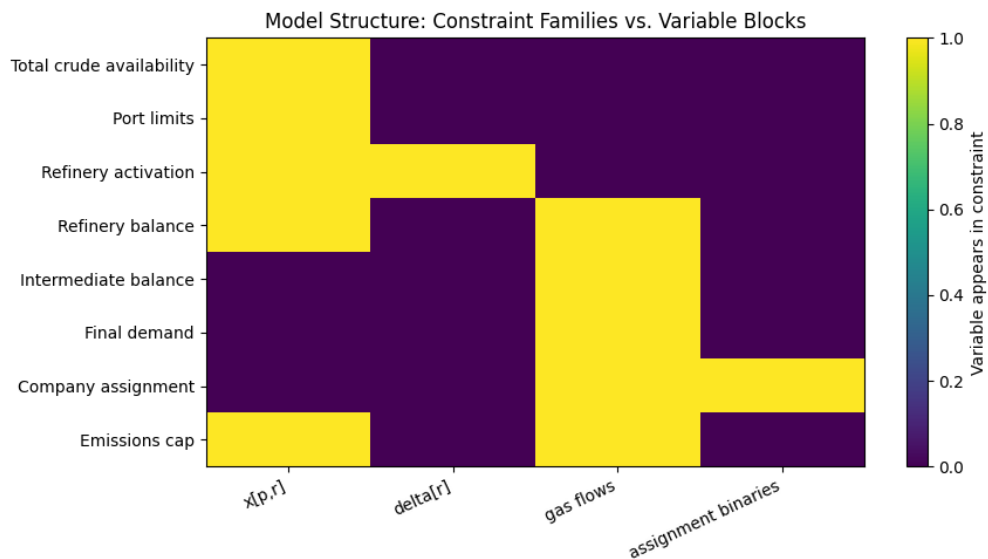


Figure 3: Model structure showing how each family of constraints interacts with each block of variables.

3. Computational Implementation

The optimization model was implemented in Python using PuLP. The notebook builds the network data, defines the variables and constraints, solves each scenario, and exports both numerical results and visual summaries. The implementation is modular: the same model-building function is reused across scenarios, with only the relevant scenario-specific constraints changed. This reduces duplicated code and makes the analysis easier to audit.

The complete source code and the notebook used to generate the results will be made available in the following GitHub repository:

[GitHub repository hyperlink](#)

The repository should include the Jupyter notebook, the requirements.txt file, the generated Excel output, and the figures used in this report. This allows the model to be reproduced without relying on screenshots of code.

4. Results

All scenarios were solved to optimality except the emissions cap case, which was infeasible under the 98% cap. The base objective value is 4,375 k€. This value is used as the benchmark for interpreting the remaining scenarios.

Scenario	Status	Objective [k€]	Δ vs. Base [k€]	Emissions	Main insight
Base case	Optimal	4,375.00	0.00	3,082.50	Benchmark solution
Refinery w shutdown	Optimal	4,450.00	+75.00	3,327.50	Small cost increase, but routes must be reorganized
Port C limit	Optimal	4,375.00	0.00	3,127.50	Same total objective, different crude allocation
Single private carrier	Optimal	5,415.00	+1,040.00	3,307.50	Most expensive feasible market structure
Route-by-route carriers	Optimal	5,365.00	+990.00	3,307.50	Flexibility saves 50 k€ versus single carrier
Company 1 bonus	Optimal	4,150.00	−225.00	3,307.50	Bonus reverses the cost ranking
Demand +10%	Optimal	4,712.50	+337.50	3,440.25	Higher demand raises both costs and emissions
Gas transport +15%	Optimal	4,710.25	+335.25	3,127.50	Cost increase without emissions increase
Emissions cap 98%	Infeasible	—	—	—	Cap is too tight for the current network

Table 6: Scenario-level results.

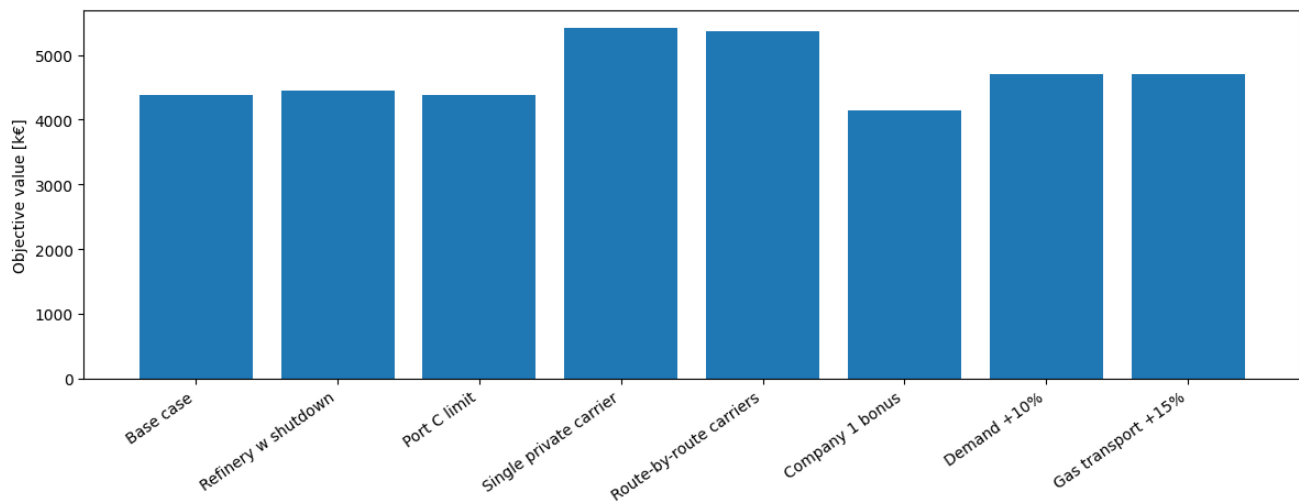


Figure 4: Objective function comparison by scenario.

The largest cost penalty comes from the private carrier scenarios. A single-carrier policy raises the objective to 5,415 k€, while allowing route-by-route assignment lowers it slightly to 5,365 k€. The 50 k€ difference is not large relative to total system cost, but it is operationally meaningful because it comes from a more flexible contracting rule rather than from additional physical capacity.

The Company 1 bonus scenario has the lowest net objective value, 4,150 k€, even though its gross gasoline transportation cost is high. This shows that market incentives can dominate pure transportation cost differences when the bonus is large enough. From the state's point of view, the bonus reduces the net cost, but it also creates dependence on a particular carrier and should be evaluated together with service reliability.

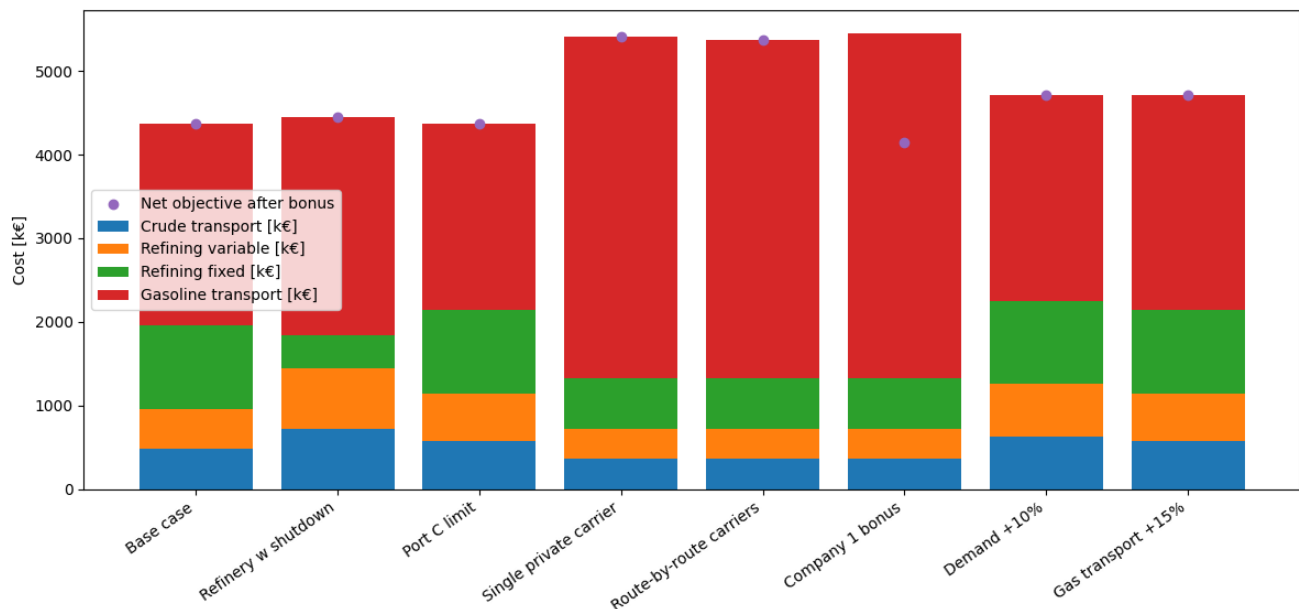


Figure 5: Cost breakdown by scenario.

The cost breakdown shows that gasoline transportation is the dominant component in most scenarios. In the base case, gasoline transportation accounts for 2,415 k€ out of a 4,375 k€ objective value. This explains why carrier assignment and gasoline transport inflation have a strong impact on total cost.

4.1 Base case and operational disruptions

In the base case, the optimal plan processes 3,600 kl of crude oil and produces 1,200 kl of gasoline. The model uses refinery u and refinery w. The main crude flows are $B \rightarrow u$ with 1,200 kl and $C \rightarrow w$ with 2,400 kl. Gasoline then moves from u to warehouse 1 and from w to warehouses 2 and 3, with warehouse 3 supplying the final demand nodes 4 and 5.

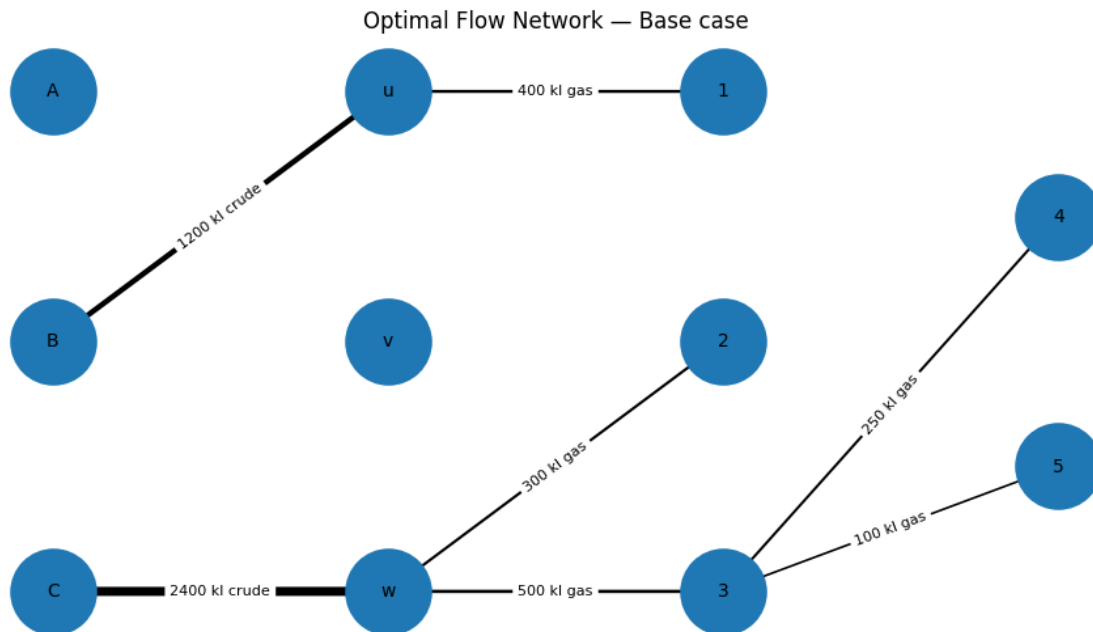


Figure 6: Optimal flow network in the base case.

When refinery w is shut down, the model remains feasible, but the network has to rely on refinery u. The crude flow becomes $B \rightarrow u$ with 3,600 kl, and refinery u supplies all gasoline demand through intermediate warehouses. The objective increases from 4,375 k€ to 4,450 k€. The increase is modest, but the operational risk is clear: the network becomes more concentrated and less resilient.

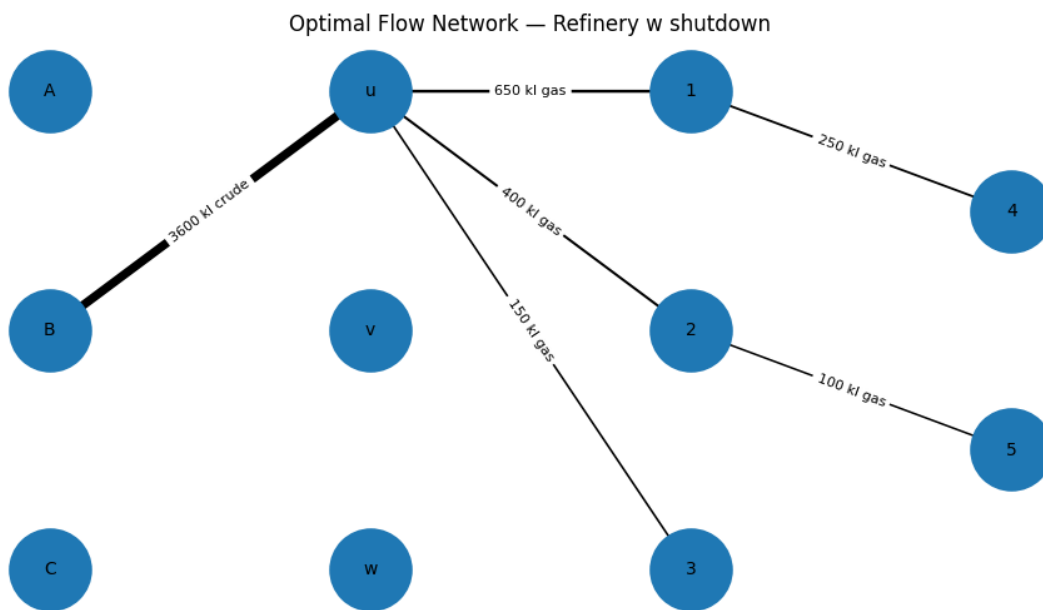


Figure 7: Optimal flow network when refinery w is unavailable.

The port C limit does not change the objective value, but it changes the allocation of crude oil. The model moves from $C \rightarrow w$ with 2,400 kl in the base case to $C \rightarrow w$ with 1,500 kl and $B \rightarrow u$ with 2,100 kl. This is a useful result: some constraints may be binding operationally without increasing the objective, because the network has enough alternative routes to absorb the restriction.

4.2 Demand and cost sensitivity

A 10% increase in demand raises the objective from 4,375 k€ to 4,712.5 k€. The network uses additional crude oil and shifts part of the production through refineries u and w. This is the scenario with the highest emissions among the feasible cases, which is expected because more gasoline has to be produced and transported.

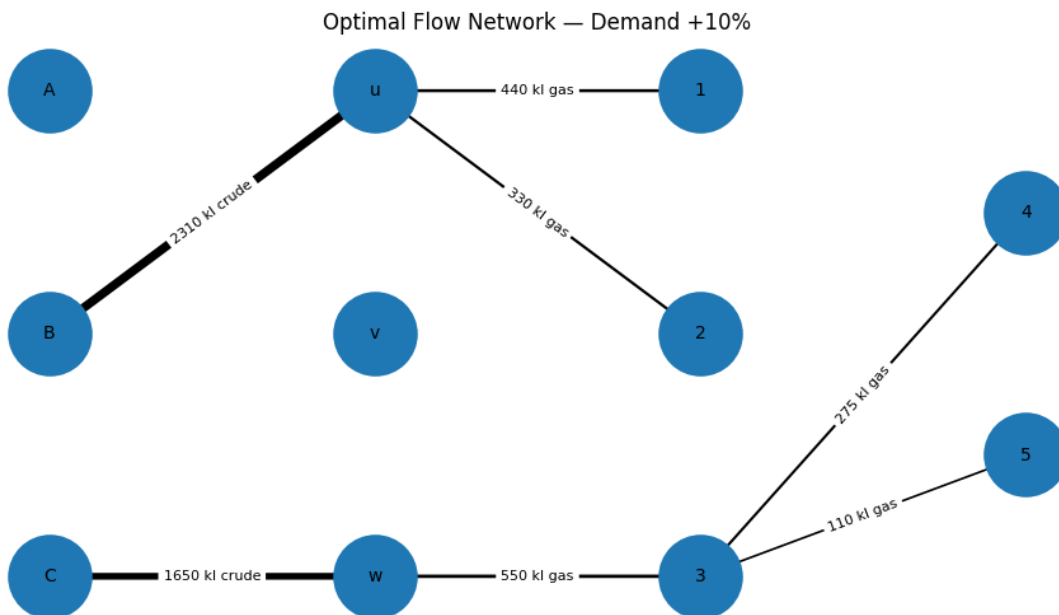


Figure 8: Optimal flow network under a 10% demand increase.

The gasoline transport cost shock increases the objective to 4,710.25 k€, nearly the same level as the 10% demand shock. However, unlike the demand shock, total emissions remain unchanged because the physical flow pattern is similar to the port C limit case. This distinction is important: a cost shock affects the objective through prices, while a demand shock affects both cost and physical system intensity.

4.3 Cost-emissions trade-off

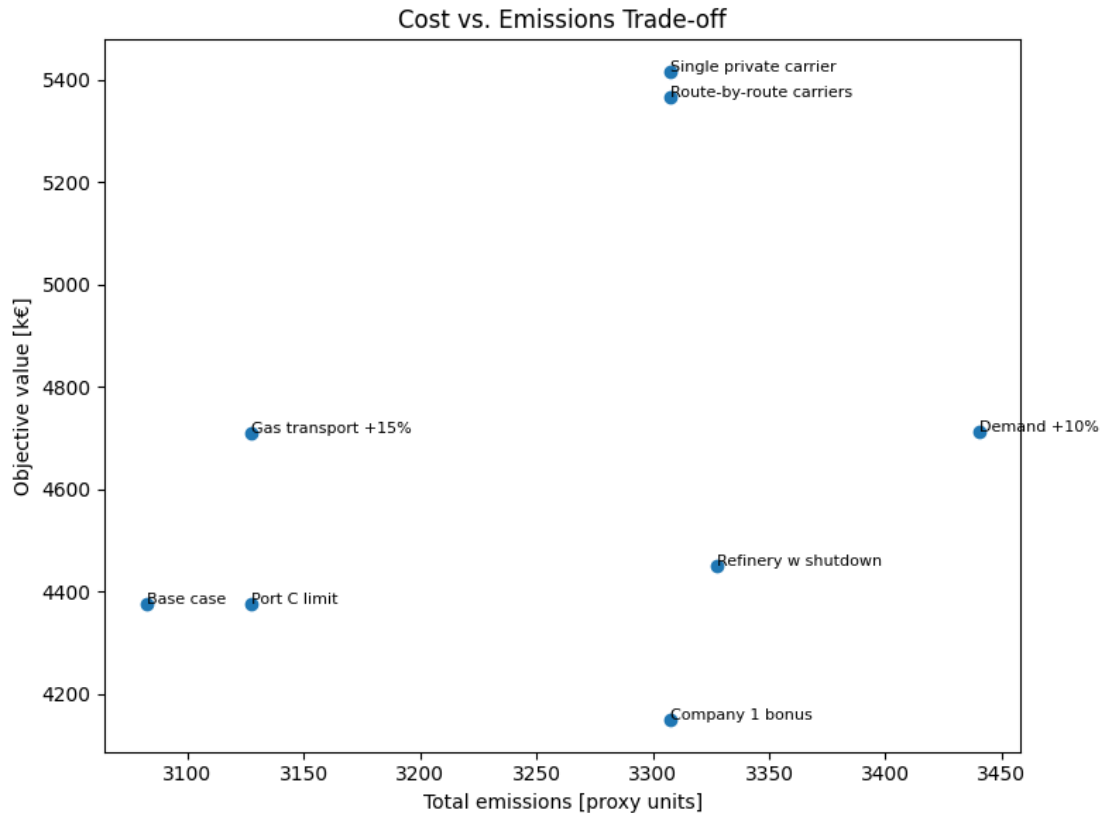


Figure 9: Cost-emissions trade-off across feasible scenarios.

The emissions analysis shows that the base case is both low-cost and relatively low-emission among the feasible scenarios. The demand shock increases emissions the most. The emissions cap at 98% of the base reference was infeasible, meaning that the current network cannot satisfy all demand while reducing emissions to that level under the available routes and technologies. This is a valuable managerial insight: sustainability targets may require new infrastructure, cleaner transportation modes, additional capacity, or relaxed demand assumptions rather than only better routing.

5. Managerial Insights

- The base network already uses a cost-efficient structure. Several restrictions change flows, but not all of them materially increase cost.
- Refinery w is important for diversification. Closing it has a limited cost effect in this dataset, but it concentrates the network around refinery u and increases operational exposure.
- Gasoline transportation is the main cost driver. Contracting decisions for gasoline carriers have a larger impact than moderate changes in crude routing.
- Route-level carrier flexibility creates savings compared with selecting a single carrier, although the savings are smaller than the overall private-carrier cost increase.

- The Company 1 bonus can reduce the net objective, but it should not be interpreted as a purely operational efficiency gain. It is a financial incentive that may come with strategic dependence.
- A strict emissions cap is infeasible with the current network. Environmental targets would require structural changes beyond route optimization.

6. Conclusion

This project applied linear optimization concepts to a realistic fuel supply chain problem. The model captures the main structure of a transshipment network: supply limits, conversion at refineries, flow conservation at intermediate warehouses, and final demand satisfaction. The resulting formulation is linear in the continuous flow variables, and it becomes a MILP when fixed refinery activation and company assignment decisions are included.

The results show that optimization is useful not only for identifying the minimum-cost distribution plan, but also for testing the effect of operational disruptions and market rules. The base case achieves an objective value of 4,375 k€. The refinery shutdown increases cost only slightly, while private carrier constraints and gasoline transport inflation have much larger effects. The demand shock raises both cost and emissions, while the emissions cap demonstrates that some sustainability targets cannot be achieved through routing alone.

Overall, the project illustrates the practical value of optimization in logistics and energy systems. A relatively compact mathematical model can provide clear guidance on routing, refinery utilization, outsourcing decisions, and the limits of the current infrastructure. The most important lesson is that the optimal solution depends not only on costs, but also on the operational and market constraints imposed on the system.

References

Bertsimas, D., & Tsitsiklis, J. N. Introduction to Linear Optimization. Athena Scientific.

Python implementation developed using PuLP for linear and mixed-integer optimization modeling.